

0.1 重积分计算

例题 0.1 设 $f(x, y)$ 在 D 上有定义, 且在某一圆盘外为 0. 求证:

$$\iint_D \left[\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \right] dx dy = 0.$$

证明 [Show/Hide Proof](#)

由题设可知 f 在 ∂D 上恒为 0, 进而

$$\frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial y}(x, y) = 0, \quad \forall (x, y) \in \partial D.$$

由 Green 公式可得

$$\begin{aligned} \iint_D \left[\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \right] dx dy &= \iint_D (f_{xx} f_{yy} - f_{xy}^2) dx dy \\ &= \iint_D [(f_{xx} f_y)_y - (f_{xy} f_y)_x] dx dy \stackrel{\text{Green 公式}}{=} \oint_{\partial D} f_{xy} f_y dx + f_{xx} f_y dy = 0. \end{aligned}$$

□

例题 0.2 设不全为 0 的实数 α, β, γ , 考虑

$$C: \begin{cases} x^2 + y^2 + z^2 = 1 \\ \alpha x + \beta y + \gamma z = 0 \end{cases}.$$

计算 $\int_C y(y-x) ds$.

解 [Show/Hide Solution](#)

记 $K \triangleq \sqrt{\alpha^2 + \beta^2}, M \triangleq \sqrt{\alpha^2 + \beta^2 + \gamma^2}$, 考虑如下正交变换

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \frac{\alpha\gamma}{KM} & \frac{\beta\gamma}{KM} & -\frac{K}{M} \\ \frac{\beta}{K} & \frac{\alpha}{K} & 0 \\ \frac{\alpha}{M} & \frac{\beta}{M} & \frac{\gamma}{M} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = T \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

于是曲线 C 方程等价于

$$\begin{cases} \left(\frac{\alpha\gamma}{KM} x' - \frac{\beta}{K} y' \right)^2 + \left(\frac{\beta\gamma}{KM} x' + \frac{\alpha}{K} y' \right)^2 + \left(-\frac{K}{M} x' \right)^2 = 1 \\ z' = 0 \end{cases} \iff \begin{cases} x'^2 + y'^2 = 1 \\ z' = 0 \end{cases}.$$

因为正交变换是等距变换, 进而保持弧长微元, 即

$$ds = \sqrt{(dx)^2 + (dy)^2 + (dz)^2} = \left| \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} \right| = \left| T \begin{pmatrix} dx' \\ dy' \\ dz' \end{pmatrix} \right| = \left| \begin{pmatrix} dx' \\ dy' \\ dz' \end{pmatrix} \right| = \sqrt{(dx')^2 + (dy')^2 + (dz')^2} = ds'.$$

又由曲线 $C: \begin{cases} x'^2 + y'^2 = 1 \\ z' = 0 \end{cases}$ 的对称性知

$$\int_{x'^2+y'^2=1, z'=0} x' y' ds' = \int_{x'^2+y'^2=1, z'=0} (-x') y' ds' \implies \int_{x'^2+y'^2=1, z'=0} x' y' ds' = 0.$$

所以

$$\begin{aligned} \int_C y(y-x) ds &= \int_{x'^2+y'^2=1, z'=0} \left(\frac{\beta\gamma}{KM} x' + \frac{\alpha}{K} y' \right) \left(\frac{(\beta-\alpha)\gamma}{KM} x' + \frac{\alpha+\beta}{K} y' \right) ds' \\ &= \int_{x'^2+y'^2=1, z'=0} \left[\frac{\beta(\beta-\alpha)\gamma^2}{K^2 M^2} x'^2 + \frac{\alpha(\alpha+\beta)}{K^2} y'^2 \right] ds' \end{aligned}$$

$$\begin{aligned} &= \int_0^{2\pi} \left[\frac{\beta(\beta - \alpha)\gamma^2}{K^2 M^2} \cos^2 \theta + \frac{\alpha(\alpha + \beta)}{K^2} \sin^2 \theta \right] d\theta \\ &= \frac{\beta(\beta - \alpha)\gamma^2}{K^2 M^2} \pi + \frac{\alpha(\alpha + \beta)}{K^2} \pi = \pi \frac{\alpha^2 + \alpha\beta + \gamma^2}{\alpha^2 + \beta^2 + \gamma^2}. \end{aligned}$$

□